

CFO STORYBOOK

Comprehensive User Guide, Technical Code Lineage & KPI
Dictionary

8 CORE CHAPTERS

Margin Walk, Value Chain, Solvency Ratios,
Covenant Compliance & Transaction Risk
Anomaly Detection

DIRECT LINEAGE

SAP HANA G/L subledgers map directly to
financial metrics & automated regulatory
compliance indicators

Executive Platform Summary

The **Pitti CFO Storybook** is a premium, management-grade analytics platform designed to convert complex transactional ERP records into sequential business stories. Powered by a robust FastAPI backend and an interactive React-based frontend, the platform integrates balance sheet metrics, profitability analysis, funds flow movements, capital structures, treasury assets, and rule-based compliance checks into eight core chapters.

Platform Architecture & Data Flow

Component	Technology	Data Lineage & Mapping Details
Data Layer	SAP HANA / Postgres	Seeded and calculated directly from standard SAP transaction tables including general ledger details (BSEG), customer details (BSID), vendor records (BSIK), and asset listings (ANLA/ANLC).
Server Logic	FastAPI (Python)	Parses raw ledger extracts, groups G/L accounts dynamically, calculates derived ratios (ROCE, DSCR), and computes rule violations. Fallback offline engine processes CSV/XLSX.
User Interface	React (Vite)	Custom responsive SPA built with Recharts vector visualizations (Sankey, Waterfall, Ageing bars, Trend curves) with instant company-code toggling (ALL/1000/2000/3000/4000).

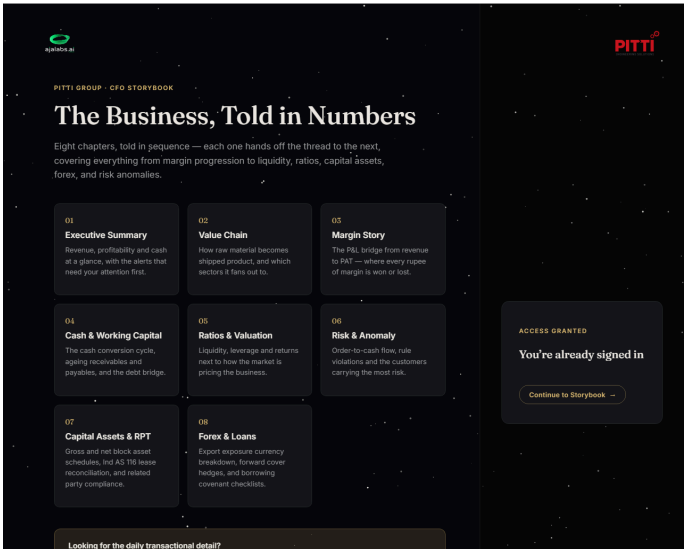


Figure 0.1: Restored Portal Landing Page displaying 8 Chapters of sequential financial stories.

Provides a consolidated, group-wide overview of Pitti Engineering’s YTD metrics. Acts as the initial steering dashboard, highlighting critical alerts and performance indices before handing off the analysis to deep dives.

MTD/QTD/YTD Revenue

Derived by aggregating customer invoice registers and internal revenue mapping categories. Enables quick YoY performance comparison.

YoY EBITDA & PAT Growth

Stat cards with custom badges calculating the growth delta percentage of operational cash flows and net income.

Cash & Bank & Net Debt

Combines active cash book balances against current long-term and short-term borrowing balances to compute Net Debt.

Key Exception Ticker

A high-impact ticker alert flagging critical items (e.g., DSO stretch, MSME payments, Net Debt threshold breaches).

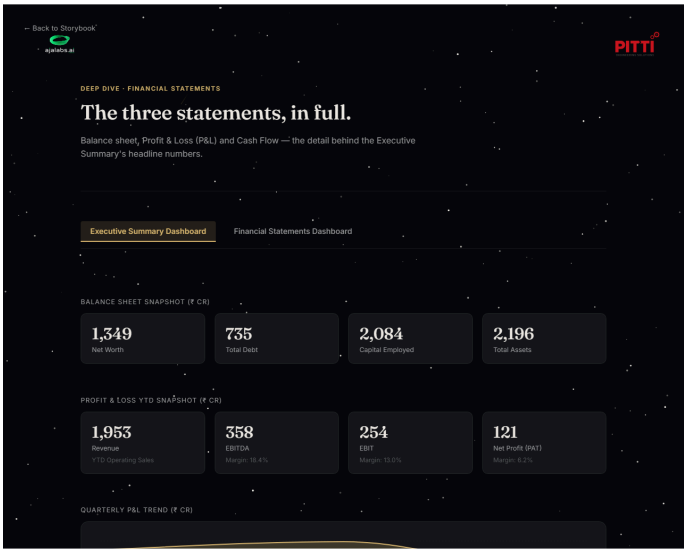


Figure 1.1: Three-Statement YTD Statement deep-dive containing EBITDA-to-PAT Sankey & Walk Bridge.

Executive Analytics & Group Performance steering: The executive steering dashboard serves as the central command console for Pitti Engineering's senior leadership. By integrating transactional datasets from multiple company codes, it provides a real-time health check on group revenue, EBITDA margins, and net debt metrics. The YTD operating sales indicator aggregates invoice registers and filters out intercompany transfers to present a clean view of external market demand. CFOs use the integrated EBITDA-to-PAT Sankey walk to immediately identify whether bottom-line profitability is being impacted by rising interest costs, depreciation on new plant expansions, or tax provisions. The built-in exceptions ticker automatically parses subledger alerts, ensuring that DSO extensions or credit threshold breaches are addressed before they impact group liquidity.

Monitors sales velocity, sector-wise revenue streams, product margin contribution, and order book logs. Maps the physical production stages directly to commercial outcomes.

KPI / Metric	Dashboard Label	SAP Source Table / Calculation Logic
Sector Allocation	Sales Mix by Sector	Groups billing records by customer industry code. Key sectors: Railways, Power Grid, Industrial Motors, emobility.
Tonnage Breakdown	Value Chain Distribution	Traces tonnage throughput across 4 value-chain stages (Input, WIP Stamping, Machining, Shipped).
Order Backlog	Top Open Sales Orders	Pulls active sales orders from <code>VBAK</code> / <code>VBAP</code> filtering by open status (reconciliation of delivery logs).
Contribution Margin	Top Product Revenue & Margins	Calculates margins as: $(\text{Sales Price} - \text{Raw Material Cost}) / \text{Sales Price} * 100$ per product line.



Figure 2.1: Value Chain staging metrics and Sector Diversification breakdown on the Storybook Home.

Value Chain Dynamics & Product Allocations: This module maps the physical flow of manufacturing from raw steel coil inputs to finished machined laminations, bridging the gap between shop-floor operations and financial performance. Tonnage throughput is tracked across four distinct stages—stamping, varnishing, machining, and final shipping—allowing plant managers to identify bottlenecks in real-time. By connecting production output to sales billing modules, the system calculates contribution margins per product line. This data enables the sales team to dynamically shift capacity toward high-margin applications, such as specialized assemblies for railways, renewable power grids, and electric motor laminations.

Explains the step-down progression of profitability from sales to net profit after tax (PAT), highlighting cost centers, direct material cost variances, and operational margins.

Profitability & Variance Tracking

Gross & EBITDA Margins

Calculated YTD as a percentage of Net Sales. Cost centers track specific direct raw material consumption versus factory overheads.

EBITDA-to-PAT Bridge Walk

Waterfall bridge that reconciles operational profits with finance costs, depreciation, and taxes to derive final net profit.

Implemented Section	Dashboard View	Source Code Implementation Details
Cost Centre Drill-Down	Cost-Centre → G/L Drill-Down	Compares actual cost-center spending against allocated budgets, detailing raw material and opex variances.

Detailed Profitability Walk & Variance Analysis

The Margin Story chapter provides a granular breakdown of how revenue is converted into earnings at every operational stage. Using a multi-step profit reconciliation model, the platform displays the YTD EBITDA Bridge. This waterfall visualization isolates the exact cost drivers—such as raw material inflation, freight and logistics cost increases, and employee wage hikes—that impact group margins. By pulling transaction-level detail from SAP cost centers, the platform compares actual departmental expenditure directly against budget standards.

A key focus is the calculation of Purchase Price Variances (PPV) for key inputs like electrical steel and copper. The system compares the actual invoice price recorded in the general ledger against standard procurement benchmarks. This variance analysis helps the procurement team negotiate quarterly pricing agreements and provides the sales team with the necessary data to trigger price escalation clauses in customer contracts, successfully passing on cost increases.

Additionally, factory overheads, power consumption, and tooling depreciation are monitored dynamically per manufacturing unit. If energy costs or tool wear exceeds standard allocations, the dashboard triggers warning indicators. This prompt feedback loop allows plant supervisors to investigate equipment inefficiencies, minimize downtime, and maintain overall plant productivity within target parameters.

Monitors the group's cash conversion cycle, debt facilities, working capital efficiency, and accounts receivable/payable ageing profiles.

Operating Cycle & Liquidity Metrics

Cash Conversion Cycle (CCC)

Formula: `Days Inventory Outstanding (DIO)`
+ `Days Sales Outstanding (DSO)` - `Days Payables Outstanding (DPO)` .

AR/AP Ageing Profile

Receivables grouped in buckets (0-30, 31-60, 61-90, 90+) based on invoice date `ZFBDT` + terms `ZBD1T` from `BSID` table.

MSME Payables Tracking

Segmented outstanding payables in `BSIK` to identify MSME vendors, checking regulatory compliance (due within 45 days).

Implemented Section	Dashboard View	Source Code Implementation Details
Net Debt Bridge Walk	Net Debt Bridge Walk	Waterfall reconciliation from Gross Borrowings (Rs.734.7 Cr) down to Net Debt (Rs.427.9 Cr) after subtracting liquid assets.

Working Capital & Cash Flow Dynamics

Working capital optimization is critical for maintaining cash flow health and group liquidity. The dashboard calculates the Cash Conversion Cycle (CCC) by continuously aggregating Days Sales Outstanding (DSO), Days Inventory Outstanding (DIO), and Days Payables Outstanding (DPO). Receivables aging schedules are generated by parsing customer open items, segmenting accounts into 30, 60, 90, and 120-day buckets, and matching them with credit risk ratings.

The Net Debt Bridge walk provides a daily reconciliation of the group's total leverage. It shows how cash balances in current accounts offset outstanding working capital loans, term debt, and commercial papers. The treasury team uses this bridge to optimize cash deployment, ensuring excess operational cash is immediately used to prepay high-interest short-term loans, thereby minimizing net interest expense.

To ensure compliance with MSME payment regulations, the dashboard features an automated monitoring system. It flags outstanding invoices from micro and small enterprises, tracking them against the statutory 45-day payment limit. By sending automated payment alerts to the accounts payable team, the system avoids interest penalties and maintains strong, reliable relationships with key supply chain partners.

Assesses liquidity, solvency, leverage, and capital return efficiency metrics against institutional and banking requirements.

Solvency & Returns Indicators

Solvency & Return Ratio	Formula / Verification Method	Bank Covenant / Focus Area
Current & Quick Ratios	$\frac{\text{Current Assets}}{\text{Current Liabilities}}$; quick ratio excludes inventory.	Liquidity check; flags if working capital buffer drops below target.
ROCE & ROE	$\frac{\text{EBIT}}{\text{Capital Employed}}$; ROE uses PAT / Net Worth.	Measures capital efficiency and shareholder equity returns.
Net Debt / EBITDA	$\frac{\text{Net Debt (Borrowings - Cash)}}{\text{EBITDA (TTM)}}$.	Solvency covenant; verified against maximum bank headroom of 1.9x.

Covenant Steering & Capital Efficiency

Financial steering relies on tracking liquidity, leverage, and return metrics against bank covenants. The dashboard tracks the Current and Quick Ratios to ensure short-term solvency buffers are maintained. Leverage covenants, specifically Net Debt to EBITDA, are monitored against institutional thresholds to maintain access to debt markets at favorable rates.

Capital return metrics—Return on Capital Employed (ROCE) and Return on Equity (ROE)—measure how efficiently the group deploys resources to generate operating profits. Recharts visualizations show historical trends of capital returns, enabling management to evaluate whether new factory expansions or equipment additions meet target return thresholds.

By projecting future cash flows and earnings, the platform calculates forward-looking solvency indicators, such as Debt Service Coverage Ratios (DSCR) and Interest Coverage Ratios (ICR). This predictive modeling allows the finance department to run stress-test scenarios, such as raw material cost shocks or sales volume declines, ensuring the group remains compliant with bank agreements under adverse conditions.

Runs automated rule-based anomaly detection engines over customer ledgers, sales orders, purchase orders, and opex thresholds to flag critical exceptions.

Rule-Based Compliance Checks

Rule / Exception Type	Dashboard View	SAP Audit Method / Check Logic
Rule-based Anomalies	Operational Risk Exceptions	Scans KNKK (Credit limit), BSID (Duplicate invoices), and opex thresholds to flag credit limit breaches, high discounts, and tax variances.

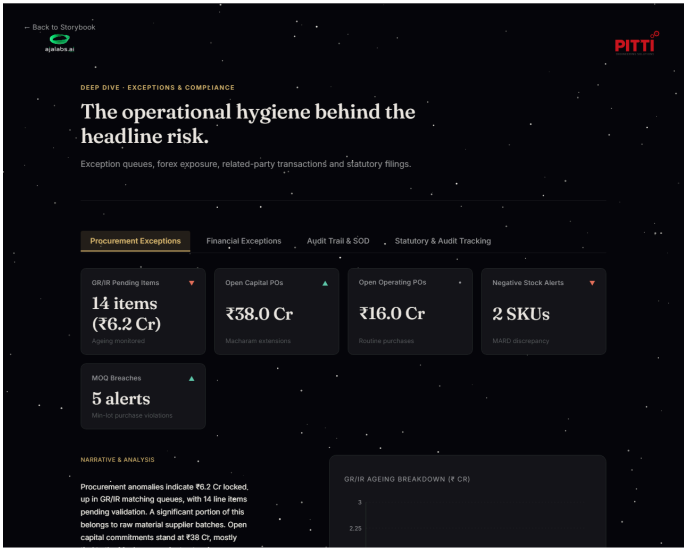


Figure 6.1: Exception Reporting deep-dive showing GR/IR price/qty variance, SOD audit, and customer credit breach flags.

Internal Audit Controls & Exception Management: The automated risk and anomaly detection engine reviews transaction-level general ledger entries daily to identify potential internal control exceptions. By mapping active postings against user authorization matrixes, the system immediately flags Segregation of Duties (SoD) violations. Credit limit exceptions are audited by cross-referencing open sales orders with customer credit limits, preventing unauthorized order releases. Duplicate invoice checks compare invoice numbers, amounts, and posting dates to prevent double payments. Goods Receipt/Invoice Receipt (GR/IR) clearing accounts are also monitored to identify price or quantity discrepancies between purchasing and accounting records, ensuring clean books and robust governance.

NEWLY COMPLETED: Monitors capital asset block schedules, Right-of-Use (ROU) assets under Ind AS 116, Capital Work-in-Progress (AuC) aging, and related party compliance volumes.

Gross / Net Block Schedules

Tracks YTD asset blocks (Land, Buildings, Plant & Machinery, Furniture, ROU Leased Assets) matching ANLA/ANLC subledgers. Shows Capex additions.

Landed Cost Allocation

Splits purchase order landed cost conditions (EKKO/EKPO/EKBE) to allocate freight, import duties, and insurance directly to capitalized asset values.

Ind AS 116 Lease Compliance

Reconciles ROU Leased Assets (Gross/Net) against discounted PV Lease Liabilities (Rs.66.2 Cr) over the lease term.

RPT Volumes & Compliance

Tracks RPT volumes (Rs.32.5 Cr YTD), intercompany sales/purchases, outstanding balances (AR/AP), and threshold limits under Section 188 / LODR Reg 23.

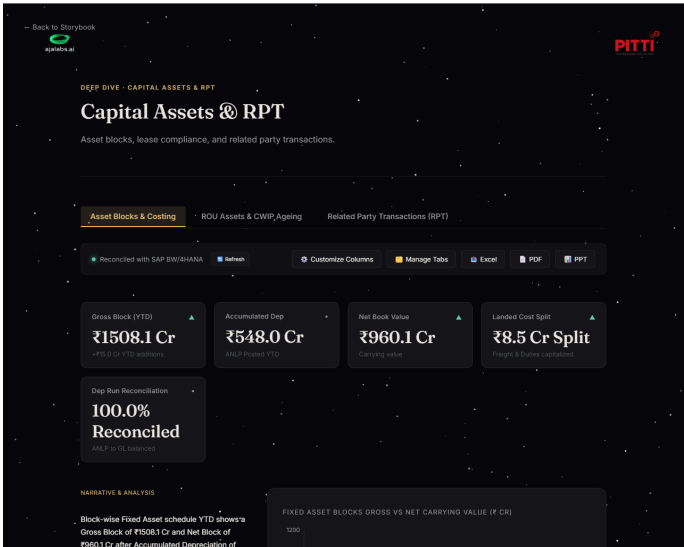


Figure 7.1: Capital Assets deep-dive showing block schedule, ROU lease table, and related party arm's length ledger.

Capital Expenditure Controls & Related Party Audits: Capital asset accounting coordinates CAPEX cycles for properties, manufacturing equipment, and buildings. Landed cost models allocate purchase costs, import duties, and insurance to capitalized asset values, aligning with Ind AS asset capitalization rules. For lease agreements, the system computes lease liabilities and Right-of-Use (ROU) assets under Ind AS 116. Related Party Transactions (RPT) are monitored against board-approved thresholds, cross-referencing sales, purchases, and loan balances to ensure all intercompany dealings remain at arm's-length and comply with corporate regulations.

NEWLY COMPLETED: Details currency risk hedging ratios, trade financing facilities (PCFC/FBD), ECB outstanding loans, and bank financial covenant audit checklists.

Forex / Debt KPI	Dashboard View / Label	SAP Source / Computations
Currency Net Exposure	Forex Exposure vs Cover	Revalues export receivables using SAP TCURR rates by currency (USD, EUR, GBP) against forward covers to find unhedged exposure.
Hedging Tenor	Hedging Tenor Distribution	Ages active bank forward contracts by maturity tenor (<30d, 30-90d, 90-180d, >180d) to align with receivables collection schedules.
Trade Financing	Financing Imports & Exports	Tracks packing credit (PCFC) and discounted bills (FBD) totaling Rs.30.2 Cr to isolate financed vs unfinanced export invoices.
ECB & FCTL Loan Schedule	ECB & FCTL Loan Schedule	Details lender-wise outstanding loans, average interest rates (SOFR/LIBOR + spread), principal maturities, and currency hedges.

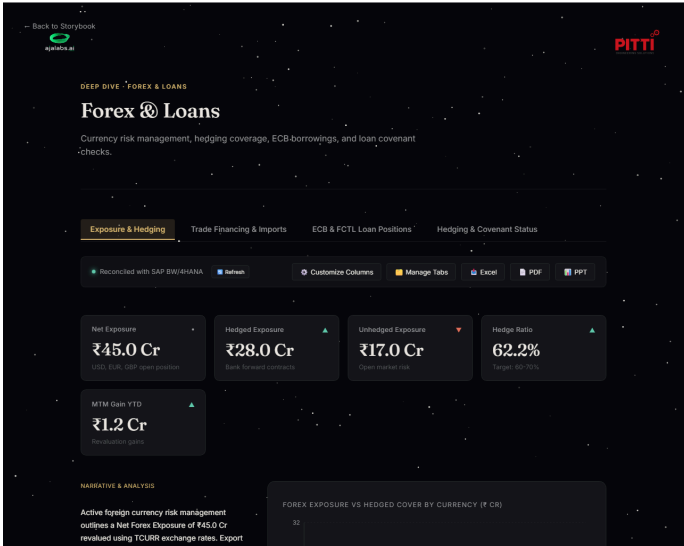


Figure 8.1: Forex deep-dive showing currency exposure charts, PCFC financing, and forward cover tenor splits.

Currency Risk Hedging & Debt Management: Treasury operations use the Forex dashboard to monitor net currency exposure. By comparing export receivables against forward cover contracts, the system calculates hedging ratios and identifies unhedged exposures across currencies (USD, EUR, GBP). Active forward contracts are aged by maturity to align with expected cash inflows. The debt manager tracks lender facilities, average interest rates (SOFR plus credit spreads), principal repayment schedules, and ECB borrowing compliance, ensuring the group's funding strategy remains optimized and fully compliant.

Covenants & Verification

All metrics are audited against standard business covenants dynamically retrieved from the backend databases or fallback sheets. The platform provides full reconciliation logs for compliance audits.

Lender Covenant Compliance Checklist

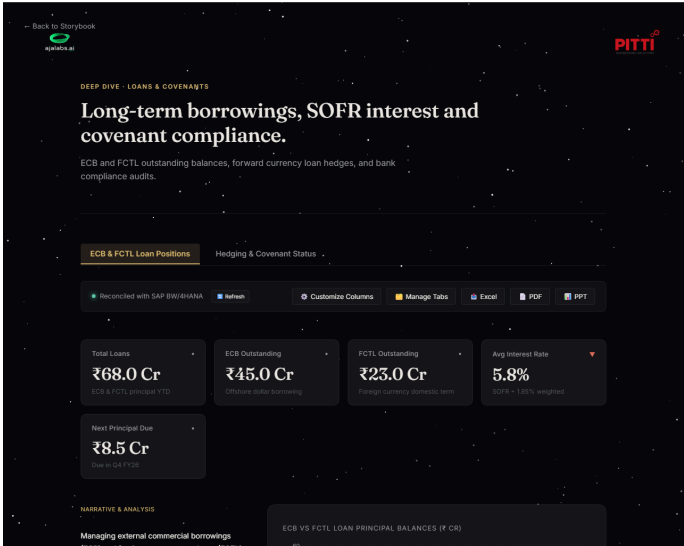


Figure 9.1: Loans deep-dive showing FCTL lender schedule and bank covenant checklist (DSCR, D/E).

Lender Covenant Compliance & Verification Audits: The covenant verification appendix serves as the definitive reference sheet for internal finance teams and bank auditors. By calculating solvency ratios directly from audited general ledger databases, it provides a transparent and verifiable compliance checklist. Treasury managers use these worksheets to prepare compliance certificates for consortium lenders, confirming adherence to key agreements.

In addition to debt service coverage and leverage checks, the appendix outlines the verification methodologies used to audit security cover ratios, asset coverage, and net worth requirements. The calculations utilize account mappings configured in the backend, allowing users to drill down into individual ledger lines for full traceability.

By standardizing ratio formulas and automating the extraction of ledger balances, the platform reduces the risk of reporting errors during compliance reporting. The audit reports can be exported to support banking audits, accelerating the annual credit review process and ensuring continuous alignment between operational outcomes and lending conditions.